Sustainable Design:

Osaka, Japan


(fcracer, 2019)

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(Osaka, 2024)

Introduction

About: Osaka, Japan

- ★ Osaka makes up a section of the second largest industrial-urban area within Japan (The Editors of Encyclopaedia Britannica, 2025).
- ★ As of July 1st, 2024, Osaka had a population of ~2.7 million (City of Osaka, n.d.a).
- ★ 92.04% of Japan's popoulation lives in urban areas, including Osaka. (*Japan - Urban Population (% of Total),* 2025).



(First Time in Osaka, n.d.)

Introduction

About: Osaka, Japan

- ★ Osaka officials consistently reference the UN's 17 Sustainable Development Goals (SDG) (Jimura, 2023).
- ★ Osaka is open to the adoption of new and innovative techniques to increase sustainability (Minami, 2021).
- ★ Major sustainable focuses include: transportation, clean energy systems, and ecological design.



(Osaka Itineraries, n.d.)

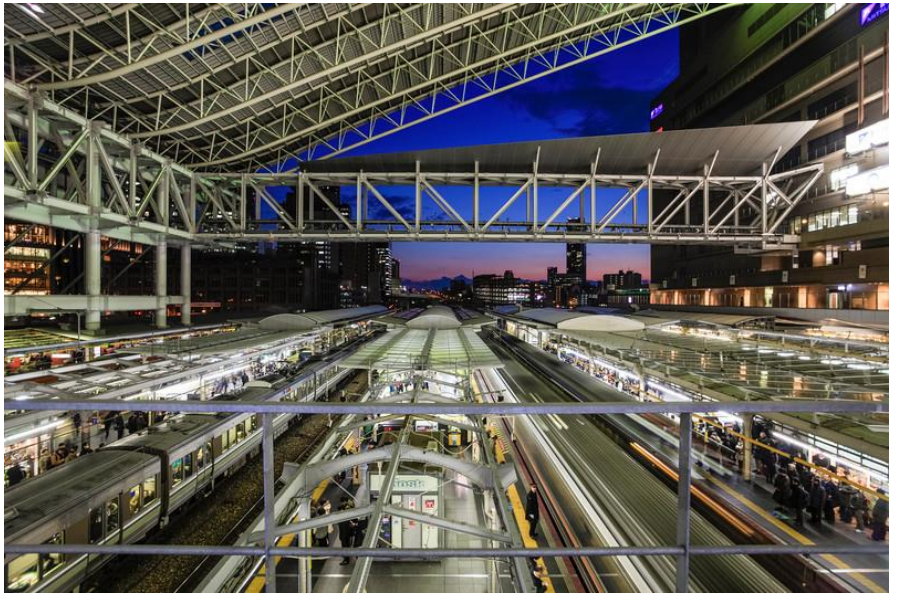
Sustainable Design:
Aspects of Interest

1. Transportation, Mobility and Travel Management
2. Decarbonization
3. Ecological Design

Sustainable Aspect #01

Osaka and Sustainable Transportation

- ★ Sustainable transit has been as a central focus in many in Asian countries, such as Japan.
- ★ Specifically, Japan has an expansive public accesses railway system.
- ★ Communities regularly access Japan's railway for
  - O work,
  - O school,
  - O and other daily activities.
- ★ Railways act as an eco-friendly solution to travel:
  - O actively reduces personal vehicle usage,
  - O carbon emissions,
  - O and additions to climate change (Jimura, 2023).
- ★ Japans transportation systems are adaptive.
- ★ Developments reference the UN's 17 SDGs (Jimura, 2023).
- ★ Due to Japan's population density, train car crowding remains an active issue.



("Getting Around Osaka," n.d.)

Sustainable Aspect #01

Osaka and Sustainable Transportation

- ★ The Osaka Municipal Transportation Bureau (OMTB) runs the Osaka urban railway system.
- ★ Osaka's urban railway system consists of:
  - O "eight rail lines [and an]..."
  - O "automated guideway transit (AGT) line" (Minami, 2021).
    - ■ Lines are regularly monitored.
- ★ Osaka consistently adopts new sustainable transit technologies.
  - O "first... variable-frequency drive train"
  - O "linear motor rapid transit line" (Minami, 2021).
- ★ Osaka seeks to provide:
  - O efficient,
  - O convenient,
  - O and accessible public transportation.
- ★ Directly limits need for personal vehicle use.



("'Along the Osaka Metro Lines' Series: New Tram," n.d.)

Sustainable Aspect #02

Osaka and Sustainable Decarbonization

- ★ “The Environmentally Advanced City of Osaka” (Fietkiewicz & Stock, 2015).
- ★ Vast amounts of energy are needed to meet population demands.
- ★ Nonrenewable energy methods have led to poor environmental conditions.
  - O Specifically, the “Heat Island” effect:
    - ■ “when the temperatures in the urban center are higher than in the suburbs,” due to air pollution from nonrenewable usage (Fietkiewicz & Stock, 2015).
- ★ Three pillars created to reduce GHG emissions:
  - O “(1) creation of a low- carbon society,”
  - O “(2) formation of a recycling oriented society,”
  - O “(3) ensuring comfortable urban environment…” (Fietkiewicz & Stock, 2015).



(Ahuja, n.d.)

Sustainable Aspect #02

Osaka and Sustainable Decarbonization

- ★ Osaka seeks to decarbonize their energy sources and promote a more sustainable energy system.
- ★ “Osaka Smart Energy Plan,” aims to “double the utilization rate of renewable energy and to improve energy efficiency” (City of Osaka, n.d.).
- ★ Specifically, Osaka is actively working towards:
  - O “making buildings ZEB-compliant when rebuilding,”
  - O “high-efficiency air conditioning,”
  - O “LED lighting in existing buildings,”
  - O “and maximizing the use of solar power generation” (City of Osaka, n.d.)
- ★ Collaboration between public and private groups.
- ★ Formation of a new renewable hydrogen energy source (City of Osaka, n.d.).
- ★ Cost and time for implementation remain as major barriers to development.



(Energy Intelligence Group, 2025)

Sustainable Aspect #03

Osaka and Sustainable Ecological Design

- ★ Osaka has been greatly impacted by the “Heat Island” effect (Masuda et al., 2020).
  - O A phenomenon where air pollution traps excess heat in urban areas, resulting in warm conditions.
- ★ Osaka created “Namba Parks…, a terraced garden…” located on “the rooftop of the 10-story shopping mall” to combat this effect (Masuda et al., 2020).
- ★ 5,300- square-meter greenery area (Masuda et al., 2020).
- ★ Ecological and social paradise for locals of the Osaka region.
- ★ Actively reduces the “Heat Island” effect through plant-based carbon sequestration (Masuda et al., 2020).



(*Namba Parks*, n.d.)

Sustainable Aspect #03

Osaka and Sustainable Ecological Design

- ★ Osaka has a densely populated landscape and expansive established infrastructure.
- ★ Grand Green Osaka & Umekita Park collaboration:
  - O aims to combine “the 11-acre Umekita Park with an 11-acre private-sector development that includes commercial and incubation facilities, universities and research institutes, a convention center, hotels, and residential units” (Grand Green Osaka, 2016).
- ★ Acts as an open space for nearby residents.
- ★ Introduces “CO2 reduction technologies” to Osaka (Grand Green Osaka, 2016).
- ★ Currently being implemented.
- ★ Will be open fully as soon as the spring of 2027, and “anticipates attracting 100 million visitors a year” (Grand Green Osaka, 2016).
- ★ Cost of implementation and collaboration between the private and public sectors remains a as major barrier.



(GGN, n.d.)

Concluding Remarks

Sustainable Osaka, Japan

- ★ Develop all sectors.
- ★ Osaka, Japan as a reference for sustainable:
  - O Transportation,
  - O Decarbonization,
  - O and Ecological Design Methods.
- ★ Long-term social and ecological sustainability.
- ★ Adaptive systems and methods.
- ★ Acknowledge barriers to implementation.
- ★ Seek stakeholder collaboration.



(GGN, n.d.)

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